Mass and Masslessness as Structural Features of Event-Density Theory

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Abstract

We show that the Event-Density (ED) primitive stack forces the *form* of mass structure — the mass operator σ_τ as a Lorentz-scalar functional of bandwidth content, the existence of a massless Case-P gauge slot, and the type-classification of mass terms by Fierz structure — while inheriting all numerical mass values, ratios, and hierarchies as empirical content. Arc M evaluates three candidate origin hypotheses for mass at primitive level (H1 fully inherited, H2 mass-as-bandwidth-signature, H3 ratios-from-taxonomy) and closes "H1-dominant with structural form-content." We establish three structural theorems. **Theorem M1** (σ_τ form): the bandwidth signature $\sigma_\tau = \hbar \cdot \sqrt{\sum_X w_\tau^X \cdot \langle(\partial_\mu \ln b_\tau^X)(\partial^\mu \ln b_\tau^X)\rangle_\tau}$ is the unique log-derivative-based Lorentz-scalar amplitude-invariant functional satisfying six selection criteria; pure amplitude rescaling cannot change σ_τ. **Theorem M2** (massless slot): the existence of at least one massless Case-P $(1/2, 1/2)$ gauge rule-type is structurally FORCED; this is initially conditional on the gauge-field-as-rule-type hypothesis (GRH) and is promoted to unconditional FORCED via Arc Q closure. **Theorem M3** (no ratio mechanism): six strong ratio claims (proportionality to representation dimension, spin, statistics class, Fierz class, bandweight pattern, individuation geometry) are systematically REFUTED; ED primitives produce no equality constraints on $\sigma_{\tau_1}/\sigma_{\tau_2}$. Mass values, mass ratios, mass hierarchies, generation count, and Higgs-mechanism choice all remain INHERITED. ED fixes the *structure* of mass, not its *values*.

1 Introduction

The Standard Model treats fermion and boson masses as empirical inputs: 19 parameters of the SM are mass values or mass ratios. No primitive-level mechanism in the SM predicts these numerically; they are fitted from data. Two long-standing puzzles persist: the *hierarchy problem* (why $m_H \ll M_{\text{Planck}}$ without fine-tuning?), and the *flavour puzzle* (why three generations with mass ratios spanning 10^5–10^{12}?).

The Event-Density (ED) program asks whether mass content is *structural* or *empirical*. Phase-1 [1] derived non-relativistic quantum mechanics from ED primitives. Arc R [2] extended this to the relativistic regime, deriving Klein–Gordon, the spin–statistics theorem, the Cl(3,1) frame, and the Dirac equation. The *form* of these equations contains a mass parameter — but the parameter's *value* is inherited. Arc M asks: is the value derivable, or only the parameter slot?

Three candidate hypotheses framed Arc M's investigation:

- **H1 (mass fully inherited):** mass values are empirical per-rule-type inputs; ED fixes only the form. *Prior plausibility: high.*
- **H2 (mass-as-bandwidth-signature):** $m_\tau = \sigma_\tau/c^2$ with σ_τ a primitive bandwidth-signature invariant. *Prior plausibility: moderate.*
- **H3 (ratios-from-taxonomy):** ED primitives constrain mass *ratios* between rule-types via taxonomy features. *Prior plausibility: low-moderate.*

Arc M closes **H1-dominant with structural form-content**: H1 substantially supported; H2 partially supported as a *form* identity; H3 substantively refuted. Three substantive structural results survive: the σ_τ master formula (Theorem M1), the existence of a structurally-forced massless slot (Theorem M2, promoted to unconditional via Arc Q back-flow), and the systematic refutation of taxonomy-based ratio claims (Theorem M3).

The motivation is methodological. *First*, demarcating clearly which mass content is structural vs. empirical eliminates ambiguity in ED's predictive scope. *Second*, the existence of a structurally-forced massless slot — ED's first "particle-class must exist" prediction — is a substantive positive result. *Third*, the structural refutation of taxonomy-based ratio claims clarifies why no SM-style mass-ratio formulae have surfaced from primitive-level analysis.

Arc M's relationship to neighbouring arcs: Arc R supplies the relativistic kinematics + Cl(3,1) frame on which Arc M builds; Arc Q (QFT extension) [3] later closes the gauge-field-as-rule-type hypothesis (GRH), retroactively promoting Arc M's massless-slot result from conditional to unconditional FORCED. The complete ED quantum sector [4] is structurally complete and UV-finite.

2 Background

2.1 Relevant ED primitives

- **Primitive 02 (chain):** worldlines γ_K. Lightlike worldlines correspond to massless rule-types.
- **Primitive 04 (bandwidth):** four-band decomposition $b_K = b_K^{\text{int}} + b_K^{\text{adj}} + b_K^{\text{env}} + b_K^{\text{com}}$.
- **Primitive 06 (four-gradient):** ∂_μ supplies Lorentz-covariant differentiation.
- **Primitive 07 (rule-type):** levers L1 (bandwidth partition w_τ^X), L2 (internal index, including Lorentz rep (j_L, j_R)), L3 (Fierz class Γ_τ), L4 (statistics Case P/R).
- **Primitive 10 (individuation):** threshold separating same-type chains.
- **Primitive 11 (commitment):** discrete events along γ_K.
- **Primitive 13 (relational timing):** proper time τ_K, degenerates on lightlike worldlines.

2.2 The σ_τ definition (preview)

The Arc M central object is the **bandwidth signature** σ_τ: a Lorentz-scalar functional of rule-type bandwidth content with units of energy. The candidate functional form, derived in Section 4, is

$$\sigma_\tau = \hbar \cdot \sqrt{\sum_X w_\tau^X \cdot \langle (\partial_\mu \ln b_\tau^X)(\partial^\mu \ln b_\tau^X) \rangle_\tau}. \tag{1}$$

The conjectured identity $m_\tau = \sigma_\tau / c^2$ connects this to the Klein–Gordon / Dirac mass parameter.

2.3 Case-P vs Case-R distinction

From Arc R [2]: rule-types divide into:

- **Case P** ($\eta = +1$, integer spin, bosonic). Mass-relevant scalar bilinear: $|\Psi|^2$.
- **Case R** ($\eta = -1$, half-integer spin, fermionic). Mass-relevant scalar bilinear: $|\bar{\Psi}\Gamma_\tau\Psi|$ with Fierz element Γ_τ.

Arc M evaluates σ_τ separately in each class. The Fierz class Γ_τ appears as a structural type-classifier for Case-R mass terms (Dirac mass for $\Gamma = \mathbb{1}$, Majorana for charge-conjugation bilinear, pseudoscalar for $\Gamma = \gamma^5$).

3 Arc M Stage M.0 — Scoping

The Arc M opening memo [5] frames the central question:

(Q-Mass) Are mass values, mass ratios, or mass hierarchies derivable from ED primitives, or are they all inherited?

The Arc M roadmap consists of four sub-stages:

- M.1.1: construct a candidate σ_τ and evaluate H2 at the form level.
- M.1.2: identify masslessness mechanisms.
- M.1.3: evaluate H3 systematically.
- M.2: synthesise a final Arc M verdict.

4 Arc M Stage M.1.1 — The Bandwidth Signature σ_τ

4.1 The construction

We seek a Lorentz-scalar, energy-dimensional, amplitude-invariant functional of bandwidth content. Six selection criteria (SC1–SC6) constrain the form:

- **SC1.** Lorentz scalar.
- **SC2.** Amplitude-invariant: shifts $b \to \alpha b$ should not change σ_τ.
- **SC3.** Dimensionally correct: units of energy via $\hbar \times$ frequency.
- **SC4.** Band-additive.
- **SC5.** Massless-permitting: $\sigma_\tau = 0$ structurally accessible.
- **SC6.** Statistics-agnostic structurally.

The candidate functional is

$$\boxed{\sigma_\tau = \hbar \cdot \sqrt{\sum_X w_\tau^X \cdot \langle (\partial_\mu \ln b_\tau^X)(\partial^\mu \ln b_\tau^X) \rangle_\tau},} \tag{2}$$

with b_τ^X interpreted as $|\Psi|^2$ in Case P and $|\bar{\Psi}\Gamma_\tau\Psi|$ in Case R.

4.2 Why the log-derivative

The choice of $\partial_\mu \ln b$ implements amplitude-invariance (SC2). For $b \to \alpha b$:

$$\partial_\mu \ln(\alpha b) = \partial_\mu \ln b, \tag{3}$$

so σ_τ is unchanged. This isolates the *spectral rate of variation* of b from its amplitude. Three alternative forms are rejected:

- **R1.** $\sigma_\tau = \hbar \cdot \langle \Box \ln b^X \rangle_\tau$ — dimensionally wrong.
- **R2.** $\sigma_\tau = \hbar \cdot \Sigma w^X \langle \partial_t \ln b^X \rangle_\tau$ — not Lorentz-invariant.
- **R3.** σ_τ with explicit amplitude dependence — violates SC2.

4.3 Status of the σ_τ form

The functional *form* in (2) is FORCED by SC1–SC6. The selection criteria are structurally motivated but not uniquely forced by primitives; we record this as a CANDIDATE *specific form*, sitting on top of a FORCED *form-class*.

4.4 Theorem M1

Theorem 4.1 (σ_τ Form, M1)**.** *The bandwidth signature σ_τ exists at primitive level as a Lorentz-scalar, amplitude-invariant, energy-dimensional functional of rule-type bandwidth content. The specific log-derivative form satisfies six structural selection criteria (SC1–SC6) and is structurally motivated. Pure amplitude rescaling $b \to \alpha b$ leaves σ_τ unchanged.*

Sketch. Construction in (2) satisfies SC1 (Lorentz-scalar via $\partial_\mu X \partial^\mu X$), SC2 (amplitude-invariance via log-derivative), SC3 (energy units via $\hbar$), SC4 (band-additive via linear $\Sigma_X w^X$), SC5 (vanishing achievable structurally), SC6 (functional form identical for Case P and R). Alternatives violating SC1–SC6 are rejected by R1–R3. □

Dependencies. Primitives 04 (bandwidth fields), 06 (four-gradient + Lorentz covariance), 07 (rule-type levers L1, L3 for Case-R Fierz), 11 (commitment for proper-time average).

5 Arc M Stage M.1.2 — Massless Rule-Types

5.1 The $\sigma_\tau = 0$ condition

From the master formula, $\sigma_\tau = 0$ iff for every band X with $w_\tau^X > 0$:

$$\langle (\partial_\mu \ln b_\tau^X)(\partial^\mu \ln b_\tau^X) \rangle_\tau = 0. \tag{4}$$

The bandwidth content has vanishing spectral rate of variation in the rest frame — the structurally massless condition.

5.2 Two distinct primitive-level realisations

Two structurally distinct mechanisms produce $\sigma_\tau = 0$:

- **MR-R (chiral masslessness, Case R).** A Case-R rule-type whose internal-index structure restricts the Dirac spinor to a single chirality (Weyl projection: $\Psi_R = 0$ or $\Psi_L = 0$). Then $\bar{\Psi}\Psi \equiv 0$ identically.
- **MR-P (gauge masslessness, Case P).** A Case-P rule-type whose L3 interface requires local gauge invariance $A_\mu \to A_\mu + \partial_\mu \alpha$. The mass term $m^2 A_\mu A^\mu$ is gauge-non-invariant; under the gauge constraint, $\sigma_\tau = 0$ structurally.

These are mutually exclusive per rule-type and tentatively exhaustive in 3+1D.

A third candidate, **MR-Λ (lightlike-only worldline)**, follows from MR-R or MR-P via the Stage R.1 Casimir $P_\mu P^\mu = m^2 c^2 = 0$.

5.3 The gauge-field-as-rule-type hypothesis (GRH)

For MR-P to apply at primitive level, the gauge connection A_μ from Stage R.1 minimal coupling must be the participation measure of a specific Case-P rule-type τ_g with internal Lorentz representation $(1/2, 1/2)$ and gauge-invariant L3. This is the **gauge-field-as-rule-type hypothesis (GRH)**. Within Arc M, GRH is a CANDIDATE; it is closed at Arc Q [3] by discharge of all five GRH refinements (R-1 through R-5).

5.4 Theorem M2

Theorem 5.1 (Massless Slot, M2)**.** *The existence of at least one massless Case-P rule-type is structurally* FORCED *at primitive level, conditional on the gauge-field-as-rule-type hypothesis GRH. Specifically, the gauge connection forced by Stage R.1 minimal coupling is identified with a participation rule-type τ_g whose gauge-invariant L3 interface produces $\sigma_{\tau_g} = 0$ via MR-P.*

Sketch. Stage R.1's local-phase invariance forces the existence of a connection A_μ. Under GRH, this connection is a participation rule-type. The MR-P mechanism requires gauge invariance to forbid the mass term $m^2 A_\mu A^\mu$, leaving $\sigma_{\tau_g} = 0$ structurally. The four GRH clauses are mutually consistent; under closure of GRH refinements at Arc Q, the hypothesis becomes unconditional FORCED. □

Dependencies. Stage R.1 (minimal coupling forcing A_μ existence), Primitive 07 L3, Stage R.2 (Lorentz representation classification), GRH (Case P statistics + rule-type identification). The five GRH refinements are R-1 lightlike-worldline reformulation (closed at Arc Q.7), R-2 gauge-quotient individuation (closed at Arc Q.2), R-3 vertex-anchored commitment (closed at Arc Q.3), R-4 non-Abelian extension (closed at Arc Q.3), R-5 vacuum/particle status (closed at Arc Q.8).

The Arc Q closure of all five refinements promotes Theorem M2 from FORCED-conditional-on-GRH to **unconditional forced**.

6 Arc M Stage M.1.3 — Mass Ratios

6.1 Six lever-evaluations

Hypothesis H3 is evaluated systematically across six structural levers:

- **L-R1 (representation dimension):** $\sigma_\tau \propto \dim(j_L, j_R)$?
- **L-R2 (spin):** $\sigma_\tau \propto s$ or $s(s+1)$?
- **L-R3 (statistics class):** Case-P vs Case-R systematically differ?
- **L-R4 (Fierz class):** Γ_τ produces specific ratios?
- **L-R5 (bandweight pattern):** w_τ^X differences produce ratios?
- **L-R6 (individuation geometry):** ℓ_{ind} differences give Compton-wavelength ratios?

6.2 Per-claim verdicts

Claim	Mechanism	Verdict
R3-A	$\sigma_\tau \propto \dim(j_L, j_R)$	REFUTED
R3-B	$\sigma_\tau \propto s$ or $s(s+1)$	REFUTED
R3-C	σ_τ ordered by P/R class	REFUTED
R3-D	σ_τ determined by Fierz class	TYPE classifier; REFUTED as ratio
R3-E	σ_τ determined by w-pattern	REFUTED as ratio
R3-F	σ_τ constrained by ℓ_{ind}	REFUTED as ratio

No strong ratio claim survives.

6.3 Why ED does not constrain ratios

σ_τ in the master formula factors into rule-type data (w_τ^X) and chain-initial-condition data ($\langle(\partial \ln b^X)^2\rangle_\tau$). Both are inherited. R.2 taxonomy provides classification but not numerical content. Mass ratios are continuous numerical quantities; ED's primitive structure produces classifications cleanly but not continuous numerical relationships.

6.4 Surviving weak orderings

Three weak inequality CANDIDATEs survive:

- **M-Order-2.** Rule-types with $w_\tau^{\text{com}} = 0$ have systematically smaller σ_τ. Inequality, not ratio.
- **M-Order-3.** σ_τ orderings inversely track individuation-threshold orderings. Dimensional only.
- **M-Fierz-1.** Fierz class structurally distinguishes mass-term TYPE. Type classification.

6.5 Theorem M3

Theorem 6.1 (No Ratio Mechanism, M3)**.** *ED primitives plus the Stage R.2 taxonomy produce no equality constraints on the ratio $\sigma_{\tau_1}/\sigma_{\tau_2}$ between two massive rule-types. Six strong-form ratio claims (R3-A through R3-F) are systematically* REFUTED*. Three weak-form orderings (M-Order-2, M-Order-3, M-Fierz-1) survive as CANDIDATE producing inequalities or type-classifications, not numerical ratios. Hypothesis H3 in equality form is rejected.*

Sketch. Each lever fails: L-R1 because representation dimension is consumed in forming the Lorentz-scalar bilinear; L-R2 because the Pauli–Lubanski Casimir relates spin to mass identically given each, but does not constrain either independently; L-R3 because Case-P and Case-R bilinears differ in sign-structure but not in magnitude scale; L-R4 because Fierz class differentiates mass-term *type* not numerical ratios within a fixed type; L-R5 and L-R6 because w_τ^X values and individuation-threshold values are rule-type data, not primitive-derived. No surviving lever produces a numerical ratio. □

Dependencies. Stage R.2 taxonomy; σ_τ master formula (Theorem M1); empirical SM data as cross-check.

7 Arc M Stage M.2 — H1-Dominant Synthesis

7.1 Verdict on three hypotheses

- **H1 (mass fully inherited): substantially supported.**
- **H2 ($m = \sigma/c^2$): provisional CANDIDATE.** σ_τ exists structurally with the right form to play this role, but the specific functional choice is not uniquely forced.
- **H3 (ratios from taxonomy): substantially refuted.**

The stage M.1.4 memo on the $m = \sigma/c^2$ identity is **declined and deferred indefinitely**.

7.2 One-line Arc M verdict

Arc M shows that ED fixes the *structure* of mass but not its *values*.

7.3 What Arc M FORCES, INHERITS, and REFUTES

FORCES: σ_τ existence (Theorem M1); amplitude-invariance via log-derivative; Case-P/R bilinear distinction; $\sigma_\tau = 0$ structurally accessible; MR-R / MR-P mutually exclusive admissible mechanisms; Statistics class determines mechanism; Fierz class fixes mass-term TYPE; massless Case-P rule-type FORCED conditional on GRH (Theorem M2; promoted to unconditional via Arc Q back-flow).

INHERITS: Numerical mass values m_τ; bandweight values w_τ^X; Fierz-class assignments Γ_τ; spectral-rate values; numerical $\hbar$, c, q; mass ratios; generation count; rule-type-to-SM-species map.

REFUTES: All six strong ratio claims (R3-A–F); no specific massless rule-type forced without GRH; no canonical MR-R/MR-P choice; no count prediction; no hierarchy mechanism; no primitive-level Higgs analogue identified at Arc M scope.

8 Back-Flow from Arc Q

8.1 GRH unconditional FORCED

Arc Q [3] closes all five GRH refinements:

- **R-1 (lightlike-worldline reformulation)** closed at Arc Q.7 via dispersion-relation argument: $\sigma_\tau = 0 \Rightarrow$ Stage R.1 Casimir $P^2 = 0 \Rightarrow$ null worldline FORCED.
- **R-2 (gauge-quotient individuation)** closed at Arc Q.2.
- **R-3 (vertex-anchored commitment)** closed at Arc Q.3.
- **R-4 (non-Abelian extension)** closed at Arc Q.3.
- **R-5 (vacuum/particle status)** closed at Arc Q.8.

GRH promotes from CANDIDATE-STRONG to **unconditional forced**.

8.2 Promotion of F-M8

Stage M.1.2 closed conditionally: *Massless Case-P rule-types: FORCED conditional on GRH* (F-M8). Post-Arc-Q.8, F-M8 promotes from conditional to **unconditional forced**:

The existence of at least one massless Case-P $(1/2, 1/2)$ gauge rule-type is structurally forced at primitive level.

This is **ED's first "particle-class must exist" prediction**.

8.3 Vacuum structure (Arc Q.8) and generation differentiation

The Arc Q.8 evaluation of mechanism G4 (vacuum-anchored generation differentiation) REFUTES this. Vacuum-sector dependence does not feed back into σ_τ at primitive level. **Generation count remains empirical.**

8.4 UV-FIN and mass renormalisation

Arc Q's UV-FIN result — primitive-level UV finiteness from Primitive 01 + 13 + 04 — has implications for mass renormalisation:

- At primitive level, mass-shift contributions from self-energy diagrams are finite.
- In the continuum approximation, standard QFT mass renormalisation produces UV-divergent integrals; UV-FIN reframes these as continuum-approximation artefacts.
- Numerical mass shifts remain INHERITED.

9 Discussion

9.1 Form vs values: the canonical Arc M framing

Arc M is the cleanest expression of ED's "form-FORCED, value-INHERITED" methodological framing. The σ_τ master formula provides a *form*; values within this form are inherited.

9.2 Relationship to Higgs-like mechanisms

Arc Q [3] evaluates five Higgs-like SSB mechanisms:

- **H1 scalar-rule-type Higgs** τ_H — ADMISSIBLE-CLEAN; matches SM phenomenology.
- **H2 bandwidth-shift via spatially-patterned condensate** — CANDIDATE.
- **H3 composite** $\bar{\Psi}\Psi$ **condensate** — ADMISSIBLE-EFFECTIVE.
- **H4 gauge-fixing artefact** — REFUTED.
- **H5 vacuum-anchored** — reduces to H1/H2.

ED admits multiple structurally-clean SSB routes; specific mechanism EMPIRICAL. H2 in naive form fails because σ_τ uses log-derivatives, immune to uniform amplitude shifts — a non-trivial structural finding.

9.3 Why ED predicts form but not values

ED's primitives admit *classifications* and *form-relations*. They do not admit *specific numerical relationships* between continuous parameters — these would require additional structural input not present in the Primitive 01–13 stack.

9.4 Implications for neutrino masses

Arc M accommodates neutrino-mass smallness as either:

- Dirac neutrinos with small Yukawa.
- Near-MR-R structure with small Majorana sector.

ED admits both structurally; neither is selected.

9.5 The hierarchy problem

Arc M does not address the hierarchy problem. The hierarchy is a numerical relationship between empirically-inherited masses, not a structural feature derivable from primitives. Arc Q's UV-FIN result reframes the cosmological-Λ divergence-form puzzle but does not solve the hierarchy puzzle for the Higgs mass.

9.6 Future work

Open questions for Phase-3:

- CC-U1 (charge quantisation from U(1) compactness).
- Schwinger-sign derivation (Arc Q.5 CANDIDATE-PLAUSIBLE).
- Primitive-level cosmological boundary conditions; SPECULATIVE.

10 Conclusion

Arc M closes the chain-mass problem at primitive level with an honest H1-dominant verdict. Three structural achievements:

- **Theorem M1 (σ_τ form):** the bandwidth signature exists structurally as a Lorentz-scalar amplitude-invariant energy-dimensional functional.
- **Theorem M2 (massless slot):** the existence of at least one massless Case-P $(1/2, 1/2)$ gauge rule-type is structurally FORCED via GRH (promoted to unconditional via Arc Q back-flow). *ED's first "particle-class must exist" prediction.*
- **Theorem M3 (no ratio mechanism):** six strong ratio claims systematically REFUTED.

Arc M sits between Arc R (relativistic kinematics) and Arc Q (QFT extension). The methodological framing — *ED fixes the structure of mass, not its values* — extends naturally into Arc Q's pattern (form-FORCED, value-INHERITED at QFT scope) and prepares Phase-3.

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